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# Early Estimation of District Paddy Yields in Sri Lanka


**Wijesinghe W. K. A. D. U.**
**269219J**



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A Machine Learning Approach Using Official Crop Statistics, Satellite Rainfall and Weather Data

Code and data github.com/Uthpalagit/paddy-yield-prediction-269219J


## Executive Summary

District paddy yields in Sri Lanka are confirmed only when the Department of Census and Statistics completes its crop-cutting survey, several months after each harvest. This project developed a model that estimates the yield of each district as soon as a season ends. A dataset of 2,276 district-seasons (1981-2026) was assembled from official crop statistics, district-averaged satellite rainfall and weather records, and subjected to extensive cleaning and validation.

Twenty modelling approaches were compared on a strictly time-based design. The selected model, an Explainable Boosting Machine, estimated yields for the unseen 2017-2026 seasons with an accuracy of 87.6% (average error 446 kg/ha, or 12.4%), against 532 kg/ha for the best naive benchmark. 86% of its district estimates fell within 20% of the official figure, and the national production it implies was within 5.8% on average outside the fertilizer crisis. Every estimate can be decomposed into the contribution of each input.

No model anticipated the collapse of 2021/22 following the fertilizer import ban, and none of the additional sources tested, including climate indices, the subsidy record, satellite crop health and soil moisture, revealed it in advance. Once one shortage season has been observed, however, a policy input enables the model to recognise the next, and the deployed model therefore includes a fertilizer-shortage scenario. The model is delivered through an interactive web application and a containerised prediction service with automated testing and drift monitoring.


## 1. Background and Objective

Rice is the staple food of Sri Lanka and is cultivated in two seasons each year. *Maha* is the main season from approximately October to March, and *Yala* is the minor season from approximately April to September. The official average yield of each district is published only after the Department of Census and Statistics (DCS) completes its crop-cutting survey. Until then, the Department of Agriculture, the Paddy Marketing Board and the relevant ministries plan procurement, storage and imports without reliable yield information. The cost of this delay became evident in the 2021/22 Maha season, when the national yield fell by approximately one third following the ban on chemical fertilizer imports, and rice had to be imported at short notice.

The objective of this project was to estimate the average paddy yield (kg/ha) of each district as soon as a season ends, using only information available at that time, namely the extent cultivated and its irrigation type, the district's own yield history, and the weather during the season. A secondary objective was to explain each estimate, so that planners can judge when to rely on it.

## 2. Data Sources

No suitable ready-made dataset was available, so the modelling dataset was constructed from public sources. Many more sources were collected than were finally used. The aim was to find the best mix of features, and every candidate was tested before it was accepted (Section 4.2). The sources are listed below.

- Department of Census and Statistics, Paddy Statistics [1]. Eight tables of average yield, sown extent, harvested extent and production for every district and season since 1978/79, with extents broken down by major irrigation, minor irrigation and rainfed cultivation.
- CHIRPS rainfall [2]. Daily satellite and rain-gauge rainfall at a resolution of approximately 5 km, averaged over the boundary of each district through the NASA/USAID ClimateSERV service, for 1981-2026.

- NASA POWER [3]. Monthly temperature, humidity, solar radiation and wind for each district.
- OpenStreetMap. District coordinates and boundaries.
- FAOSTAT [4] and NASA MODIS. National fertilizer imports and satellite vegetation greenness (NDVI). Both were evaluated as additional inputs (Section 4.2) but were not retained.
- NOAA climate indices and crop health [12, 13]. The El Niño (ONI) and Indian Ocean Dipole (DMI) indices, and weekly vegetation health indices averaged over the rice-growing area of each district (1982-2026). These were also evaluated and not retained.
- Government fertilizer policy [14, 15]. The subsidy regime in force in each season since 1981, from the published policy record, including the import ban of May 2021 and the shortage of 2022.
- ERA5-Land soil moisture [16]. Daily root-zone soil moisture for each district from 1980, used with NASA POWER to derive evapotranspiration and stand-ins for reservoir storage and heat stress. These were also evaluated and not retained.
- Provinces and agro-ecological zones. The province and the Department of Agriculture climate zone of each district, evaluated as qualitative features and not retained (Section 4.3).

The unit of analysis is the district-season. The 27 units comprise the 25 administrative districts and two irrigation project areas (Mahaweli System H and Udawalawe) that the DCS reports separately. The target is the average yield in kg/ha, and the final dataset contains 2,276 district-seasons from 1981 to 2026. The 23 input features of the final model are listed in Table 1 (Section 4.2).

All the data is publicly released and aggregated to district level. No personal or sensitive information about farmers or households was collected or used, so no consent was required, and each source is cited and used within its terms of release.

## 3. Data Preparation and Quality Assurance

The official tables required substantial cleaning. Every issue was identified through an explicit check, corrected, and the correction verified before proceeding. The principal findings are listed below.

- Inconsistent structure. The header row and column differed between files, and irrigation categories were spelt inconsistently. A cell count confirmed that the reshaping retained all 34,646 data cells.
- Inconsistent district names. 34 spellings were found for 27 units, together with a national total and a separate highland category, both of which were removed.
- Mislabelled seasons. The 2020/21 Maha column was labelled "2000/2021" in three tables. Read as recorded, it would have overwritten the 2000/01 season. A year recorded as "198" was corrected to 1980.
- Impossible values. 21 yields were recorded as exactly zero, almost all in Yala seasons of 1988-1994 in the Northern and Eastern provinces, although the same records show up to 37,000 ha harvested. These reflect seasons in which the survey could not be conducted during the conflict and were treated as missing. One harvested extent exceeded the sown extent by 87% and two irrigation breakdowns did not sum to the total. These values were also set to missing.
- Inconsistent tables. The production table did not reconcile with harvested extent multiplied by yield and was not used, as it is in any case derived from the target.
- Weather resolution. NASA POWER provided only 19 distinct weather series for the 27 units, because neighbouring districts fall in the same grid cell. This finding led to the adoption of district-averaged CHIRPS rainfall, which provides a distinct series for every unit.
- Outliers. Extreme but genuine seasons, such as the 2021/22 collapse and record yields in irrigated areas, were retained. Six yields measured on less than 100 ha were excluded from model training.
- Placeholders, duplicates and types. 6,130 cells recorded as a dash and the NASA code -999 were converted to missing values. No duplicate district-seasons were found, and every column was converted to its proper type.

Weather was aggregated over each season's growing window according to the Department of Agriculture crop calendar. Because CHIRPS is daily, two further indicators could be derived, namely the longest dry spell during the growing months and the number of days with at least 50 mm of rain. History features (the previous same-season yield, the three-season average, the previous season and an ARIMA trend forecast) were computed only from earlier seasons to prevent information leakage.

The remaining 3,372 gaps in the inputs were filled with the median of the training seasons inside the modelling pipeline, never on the whole table, and a missing yield was never filled. For the models that require it, numeric inputs were standardised and district and season were one-hot encoded, while CatBoost and the EBM used the cleaned columns natively. A log transform of five skewed features (358 against 355 kg/ha) and a robust scaler (355.2 against 355.0) gave no gain, and correlation and VIF checks led to one of three temperature measures being kept.

Exploratory analysis showed that the rainfed share is the strongest single correlate of yield ($r = -0.61$, against -0.26 for heavy-rain days), that Maha exceeds Yala in 65% of district-years (by 136 kg/ha on average, $p < 0.001$), and that the yield distribution is close to symmetric (skew 0.4), so the target was modelled without transformation.

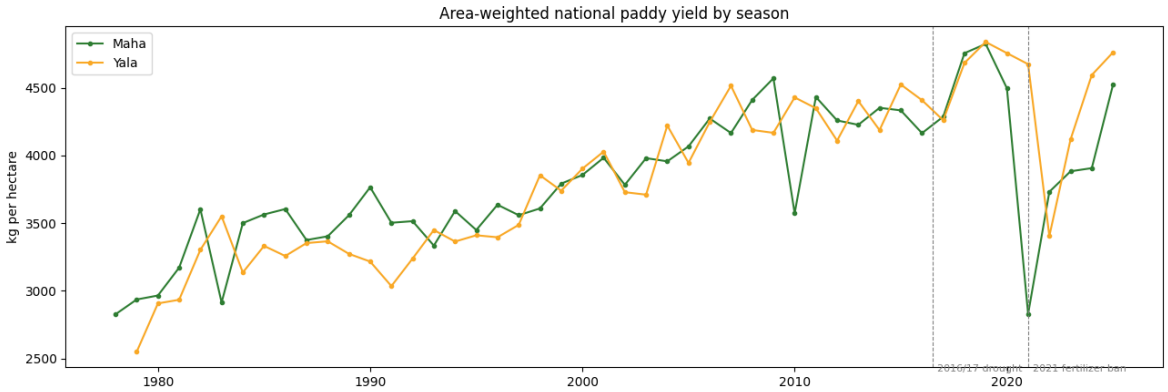



*Figure 1. Area-weighted national yield by season, showing the long-term increase and the 2021/22 collapse to 2,828 kg/ha.*

## 4. Methodology

### 4.1 Validation design

Because the data are ordered in time, a random split would allow the model to learn from the future. The data were therefore split by time. Models were trained on 1981-2016 and evaluated once on the 446 district-seasons from Yala 2017 to Maha 2025/26. All modelling decisions were made on seven yearly validation folds within the training period (training on all seasons before a given year and validating on that year, for 2010 to 2016). Preprocessing (median imputation, standard scaling and one-hot encoding of district and season) was performed inside pipelines fitted on training data only. CatBoost and the EBM, which handle categories and missing values themselves, received the cleaned columns directly, and training them through the same pipeline changed their validation error by less than 1%. The most recent Polonnaruwa season was held out entirely for a final check, and nine leakage checks passed in code.

### 4.2 Feature selection

Feature selection determined which inputs the model should use. Each candidate addition was tested individually with three different algorithms, held at fixed settings. Improvements to data quality were adopted where they did not increase the average validation error, while new external data sources were required to reduce it by more than 1%, consistent with the threshold used for model selection. On this basis, CHIRPS rainfall, the dry-spell and heavy-rain indicators, and a single temperature variable were retained. Nitrogen fertilizer imports, conditions in neighbouring districts, satellite greenness, the El Niño and Indian Ocean Dipole indices (+0.3% and -0.6%), the fertilizer subsidy level (-0.2%), crop health over the rice fields (-0.8%) and an alternative target formulation were not retained. Fertilizer imports were of particular interest, but in the training

years import volumes fluctuated widely without corresponding changes in yield, so the 2021 reduction did not appear unusual to the model. Satellite crop health was equally uninformative about the crisis. In Maha 2021/22 the peak greenness of the rice fields (NDVI 0.400) was within the normal range of the decade, although yields fell to 61% of their three-season average.

Further candidates were tested after the main analysis was complete. These were soil moisture and evapotranspiration (+0.2%), stand-ins for reservoir storage and planting conditions (+0.6%), heat in the hottest month (0.0%) and weather at each growth stage, namely heat at flowering, night temperature and sunshine during ripening, and rain at harvest (-1.5%), and four qualitative features, namely the fertilizer policy regime, province, climate zone and El Niño phase (0.0%). The harvested share was also scored, only to illustrate leakage. It appeared 2.7% better because it is published together with the yield.

The final feature set is listed in Table 1. It contains 21 numeric features together with the district and the season.

| Group | Features | Description |
|---|---|---|
| Target | yield_kg_ha | Average yield of the district-season (kg/ha) from the DCS crop-cutting survey |
| Location and season (2) | district, season | The 27 production units, and Maha or Yala |
| Yield history (5) | yield_same_season_lag1, yield_same_season_roll3, yield_prev_season, national_prev_season, arima_forecast | The same season last year, the average of the last three same seasons, the preceding season, the area-weighted national yield of the previous season, and an ARIMA(1,1,1) forecast from earlier seasons |
| Extent and irrigation (5) | sown_ha, share_major, share_minor, share_rainfed, sown_vs_prev3 | The extent sown, its shares under major irrigation, minor tanks and rainfed cultivation, and the extent sown compared with the previous three same seasons |
| Rainfall from CHIRPS (7) | rain_season, rain_anomaly_pct, rain_wettest_month, rain_driest_month, rain_before, dry_spell_days, heavy_rain_days | Total rain in the growing window and its departure from the 1981-2010 normal (%), rain in the wettest and driest months, rain before the season, the longest run of dry days and the number of days with 50 mm or more |
| Other weather from NASA POWER (4) | tmax, rh, solar, wind | Mean daily maximum temperature, relative humidity, solar radiation and wind speed in the growing window |

*Table 1. The target and the 23 input features of the final model. The deployed model adds a yes/no fertilizer-shortage input for scenario estimates (Section 5.4).*

### 4.3 Further feature checks

To test whether some other combination of all 43 candidates would perform better, three standard feature selection methods were applied, namely Lasso [17], permutation importance [18] and backward elimination [19]. To stop the methods grading their own choice, they selected features on the 2010-2013 validation years and were judged on 2014-2016. These late checks, however, reused validation years that had already informed every earlier decision, and with enough attempts some options look better on the same years by chance, which is known as overfitting to the validation set. They were therefore reported but not permitted to change the model. On the test seasons, neither the growth-stage weather (447 kg/ha) nor the consensus of the three methods (464 kg/ha) improved on the final feature set (446 kg/ha), which supports that decision. A blend with ridge regression, weighting recent seasons and tuning the EBM each changed the error by about 1% or less, and principal component analysis of the weather features made it worse (by 0.2 to 1.8%). The qualitative features were the one addition that improved the EBM on both validation (341 against 347 kg/ha) and test (434 against 446 kg/ha), but the gain was small and unstable, so it is recommended for confirmation on the 2026/27 seasons rather than adopted. Of the eleven datasets loaded, the final model uses five, namely the DCS statistics, CHIRPS rainfall, NASA POWER weather, the fertilizer policy record and OpenStreetMap to locate each district. The other six were dropped completely.

The districts were also grouped without labels by K-means clustering [20], using six profile features from the training seasons, namely average yield, the swing of yields around the three-season average, the rainfed and

major-irrigation shares, season rainfall and sown extent. The elbow method and the silhouette score [21] indicated three clusters. These were eleven major irrigation districts of the dry zone with the highest yields, nine high-rainfall districts of the wet zone and hill country with the lowest yields, and seven northern and eastern districts whose yields swing about twice as much as elsewhere. Found without any labels, this grouping reproduces the familiar agro-ecological division and coincides with the districts where the model is least accurate. As a feature of the EBM, the cluster changed the validation error by between -0.4% and +1.1% across four random seeds, no more than the variation between seeds, so it was not adopted. Ward hierarchical clustering [26] produced nearly the same three groups (adjusted Rand index [27] 0.80), so the grouping does not depend on the algorithm.

### 4.4 Model selection and parameter engineering

Feature selection fixed the inputs before any algorithm was chosen, and every model was then compared on the same inputs and the same validation folds. Keeping the two steps apart keeps the number of combinations small and limits overfitting to the validation years. Twenty candidate approaches were compared. They comprised naive and ARIMA baselines, ridge regression and a decision tree, bagging and random forest, XGBoost, LightGBM and CatBoost, a stacking ensemble, two hybrids (ARIMA with a random forest on its residuals, and a ridge and CatBoost blend), and seven additional methods. Each family was included for a reason tied to the data, and no default settings were used. With about 1,800 training rows, a strong signal in a few history features and weak, noisy weather effects, the tree models were deliberately kept shallow with large leaves, so that they learn broad patterns rather than individual district-seasons.

- Baselines. The same season last year, the three-season average and an ARIMA(1,1,1) forecast for each district set the level that any learned model must beat, since yields change slowly from season to season.
- Linear models. Ridge regression applies an L2 penalty that stabilises the correlated history features and, unlike trees, can extrapolate beyond the training range. Huber regression, which limits the influence of extreme seasons and whose penalty was tuned on the validation years, and a linear mixed-effects model with a separate baseline for each district were added as robust and panel-data variants.
- A single decision tree, the building block of the ensembles, shows what combining trees adds. Its depth and leaf size were tuned, and cost-complexity pruning was also tried but did not improve it (391 against 390 kg/ha).
- Bagging and random forest. Both average 500 trees with a maximum depth of 8 and at least 10 district-seasons in each leaf, which reduces the high variance of a single tree. The random forest also considers one third of the features at each split (max_features of 1/3, the usual rule for regression) and samples 80% of the rows for each tree, so that the trees are less alike.
- Boosting. XGBoost, LightGBM and CatBoost capture non-linear weather effects and interactions. All three used a learning rate of 0.03 and shallow trees (depth 3 for XGBoost, 8 leaves for LightGBM and depth 4 for CatBoost) with strong L2 penalties (5 for XGBoost and LightGBM, 10 for CatBoost). XGBoost and LightGBM also used 80% of the rows and 70% of the columns for each tree. The number of trees was set by early stopping on the 2015-2016 seasons, at 359 for XGBoost, 248 for LightGBM and 736 for CatBoost. CatBoost was included because it encodes the 27-level district column natively. XGBoost's minimum split gain (gamma) was also tested and made no difference.
- Stacking and hybrids. A stacking ensemble of ridge, random forest, LightGBM and CatBoost with a ridge meta-model trained only on forecasts of each year made from earlier years, ARIMA with a random forest on its residuals, and an equal blend of ridge and CatBoost test whether combining a linear trend with non-linear effects helps.
- Methods not covered in the module. The EBM [7] (Section 4.5), NGBoost [8], which predicts a full probability distribution, Gaussian process regression for smooth extrapolation, extremely randomised trees for noisy data, and histogram gradient boosting as a further boosting implementation.

The designed settings were checked rather than assumed (Figure 2). The random forest's validation error barely changed beyond about 200 trees, so 500 trees lie past the point of diminishing returns, and its training error stayed near 253 kg/ha, showing that the shallow trees did not memorise the training seasons. For XGBoost, the training error kept falling with more rounds while the validation error reached its minimum near the early-stopping point and then rose, which is the sign of overfitting that early stopping prevents. A grid search around the LightGBM and CatBoost settings lowered the validation error by only 3-6 kg/ha, well within the year-to-year spread of 84-90 kg/ha, so the more regularised designed values were kept. A randomised search was then conducted as a sensitivity analysis of the designed configurations, drawing 25 XGBoost, 20 random forest and 12 EBM settings from wide ranges and evaluating each on the same seven validation years. The best configurations reduced the validation error by 6.4, 8.4 and 4.9 kg/ha respectively. These differences are small relative to the fold-to-fold standard deviation of 79-87 kg/ha, and the minimum of several noisy validation scores is known to overstate the true improvement, particularly when the same years have informed earlier decisions. Under the adoption criterion applied to LightGBM and CatBoost, the designed configurations were therefore retained. The designed settings lay within the near-optimal region of each search, whereas poorly chosen settings increased the error by up to 67 kg/ha, and the best tuned EBM (341.7 kg/ha) remained more accurate than the best tuned XGBoost (343.7) and random forest (359.8), so the choice of model is robust to tuning. The search indicated that a smaller EBM learning rate may yield a modest improvement, which is noted for evaluation when the model is retrained on new seasons. Two methods from published Sri Lankan studies [5, 6] were rebuilt separately as a benchmark (Section 6). They were not candidates for selection.

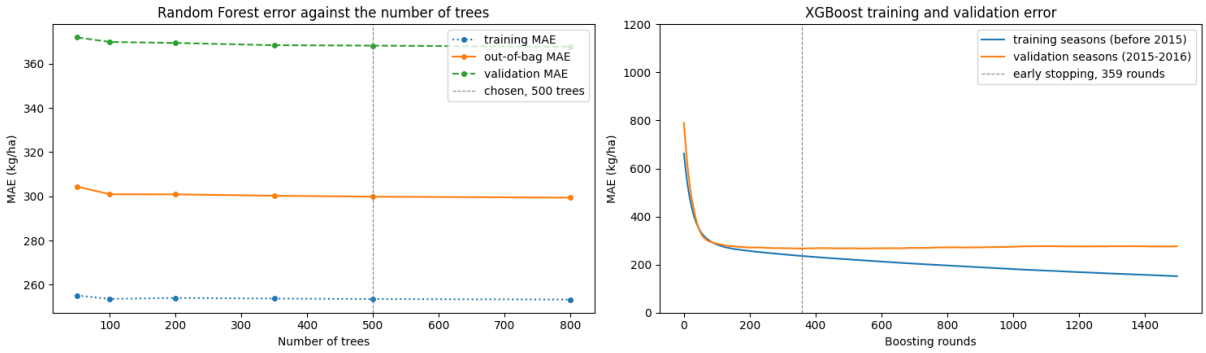



*Figure 2. Random forest error against the number of trees (left) and XGBoost training and validation error by boosting round, with the early-stopping point (right).*

The selection rule, applied to validation results only, was the lowest mean absolute error (MAE). Models within 1% of the best were treated as equivalent, given the year-to-year variation of around 80-90 kg/ha, and the most stable of these was selected.

### 4.5 The selected algorithm, the Explainable Boosting Machine

The EBM was not covered in the module. It is a generalised additive model, which means that each estimate is the sum of a baseline and one learned function of each feature, together with a few functions of two features. The functions are learned by boosting. The algorithm fits a long sequence of very small trees to the remaining error, each with a single split on only one feature, using a small learning rate (0.04) and stopping early on an internal 15% validation sample. The trees for each feature are then added together into one curve, called a shape function, which is stored as a lookup table. The final model averages eight such fits on resampled data and includes five pairwise interactions.

This design differs from the standard models in important ways. A decision tree combines many features in a chain of splits, so an estimate depends on a long sequence of conditions, and a single tree is unstable (390 kg/ha on validation, the weakest learned model). Linear and logistic regression are also additive, but each effect is forced into a straight line or a fixed curve, so they cannot represent a threshold such as the heavy-rain penalty. k-nearest neighbours predicts from the most similar past cases, has no explicit effect for each feature and depends heavily on how the inputs are scaled. Gradient boosting methods such as XGBoost grow trees over all features together, which captures interactions but requires SHAP to explain. The EBM keeps the flexibility of

boosting while every effect can be plotted and read exactly.

It was selected for three reasons. It tied with the most accurate model on the validation years and had the steadiest error between years. Its explanation is exact rather than approximate, which matters when a planner must decide whether to trust an estimate. It also handles the district and season categories and missing values without encoding or imputation.

## 5. Results

### 5.1 Model comparison

| Model | Family | Valid. MAE | Test MAE | Test RMSE | Test MAPE (%) | Test $R^2$ | Test $R^2$, excl. 2021/22 shock |
|---|---|---|---|---|---|---|---|
| Naive, same season last year | Baseline | 438 | 561 | 750 | 15.5 | 0.38 | |
| Naive, 3-season average | Baseline | 366 | 532 | 720 | 14.8 | 0.43 | |
| ARIMA(1,1,1) per district | Baseline | 375 | 509 | 687 | 14.2 | 0.48 | |
| Ridge regression | Weak learner | 353 | 438 | 620 | 12.3 | 0.58 | 0.74 |
| Decision tree | Weak learner | 390 | 533 | 717 | 15.0 | 0.43 | 0.64 |
| Bagging (500 trees) | Bagging | 370 | 482 | 673 | 13.6 | 0.50 | 0.71 |
| Random Forest | Bagging | 368 | 465 | 640 | 13.0 | 0.55 | 0.72 |
| XGBoost | Boosting | 350 | 455 | 634 | 12.8 | 0.56 | 0.72 |
| LightGBM | Boosting | 357 | 463 | 640 | 13.0 | 0.55 | 0.71 |
| CatBoost | Boosting | 353 | 456 | 624 | 12.7 | 0.57 | 0.71 |
| Stacking (Ridge, RF, LightGBM, CatBoost) | Stacking | 360 | 443 | 624 | 12.7 | 0.57 | 0.74 |
| Ridge + CatBoost blend | Hybrid | 348 | 443 | 617 | 12.4 | 0.58 | 0.73 |
| ARIMA + Random Forest residuals | Hybrid | 369 | 488 | 667 | 13.8 | 0.51 | 0.70 |
| Explainable Boosting Machine (selected) | Not in lectures | **347** | **446** | **616** | **12.4** | **0.58** | **0.72** |
| NGBoost | Not in lectures | 343 | 459 | 645 | 13.0 | 0.54 | 0.72 |
| Linear mixed-effects | Not in lectures | 355 | 441 | 626 | 12.4 | 0.57 | 0.75 |
| **Model** | **Family** | **Valid. MAE** | **Test MAE** | **Test RMSE** | **Test MAPE (%)** | **Test $R^2$** | **Test $R^2$, excl. 2021/22 shock** |
| Gaussian Process | Not in lectures | 361 | 426 | 602 | 12.0 | 0.60 | 0.74 |
| Histogram Gradient Boosting | Not in lectures | 355 | 458 | 634 | 12.9 | 0.56 | 0.71 |
| Extra Trees | Not in lectures | 361 | 467 | 632 | 12.9 | 0.56 | 0.71 |
| Huber regression | Not in lectures | 354 | 443 | 626 | 12.4 | 0.57 | 0.74 |
| Stacked XGBoost [5] | Published | 345 | 451 | 630 | 12.8 | 0.56 | 0.73 |
| Support vector regression [6] | Published | 388 | 443 | 609 | 12.2 | 0.59 | 0.70 |

*Table 2. Validation (2010-2016) and test (2017-2026) results. MAE and RMSE in kg/ha. The two published methods are a benchmark, not candidates.*

- NGBoost achieved the lowest validation error (343 kg/ha), and the EBM (347 kg/ha) fell within the 1% equivalence band. The EBM was selected as the more stable of the two (standard deviation across years of 79 against 90 kg/ha).

- On the test seasons, the EBM achieved an MAE of 446 kg/ha (12.4%), an RMSE of 616 kg/ha and an $R^2$ of 0.58, a clear improvement on the naive three-season average (532 kg/ha) and ARIMA (509 kg/ha). A Wilcoxon signed-rank test found no significant difference between the EBM and ridge regression, the strongest simple model on test ($p = 0.14$), and a highly significant improvement over the naive baseline ($p < 0.001$).
- Overfitting was checked directly. The EBM's error was 235 kg/ha on training, 347 on validation and 446 on test, early stopping halted XGBoost where its validation error began to rise, and the rise on test affected every model. Half of the EBM's estimates were within 328 kg/ha (median absolute error) and its bias was -16 kg/ha. A Diebold-Mariano test on the 18 seasons confirmed the gain over the naive average ($p = 0.025$) and no significant difference from ridge regression ($p = 0.61$).
- The tree ensembles were competitive but not better. XGBoost (350 kg/ha), CatBoost (353), the random forest (368) and bagging (370) all had a higher validation error than the EBM, and on test they fell further behind (455-482 kg/ha against 446), because a tree cannot predict yields outside the range seen in training. The single decision tree was the weakest learned model (390 kg/ha). The ridge and CatBoost blend (348) came close, but added complexity without a clear gain and is harder to explain. Stacking reached only 360 kg/ha, with the least stable error of all models between years, once its meta-model learned only from forecasts of earlier years.
- NGBoost had the lowest validation error, but its error varied most between years, it fell to 459 kg/ha on test, and its own 80% intervals held only 46% of the time (Section 5.5). Ridge regression was slightly better on test (438 kg/ha), but it was worse on validation (353), the difference was not significant ($p = 0.61$), and choosing it after seeing the test results would bias the evaluation. The EBM was therefore the best balance of accuracy, stability and explanation.
- Models with smooth or linear components (Gaussian process, ridge, Huber and mixed-effects regression) performed slightly better on the test period than tree-based models. The test period contained record yields in 2018-2020 followed by a collapse, which requires extrapolation beyond the range observed in training. The deployed model was not changed in light of these results, as selecting on the test set would invalidate it as an independent evaluation.
- The residuals of the EBM showed mild autocorrelation from one season to the next (median Durbin-Watson [28] statistic 1.68 across districts) and almost constant spread, which is why the Diebold-Mariano test averaged the errors per season.

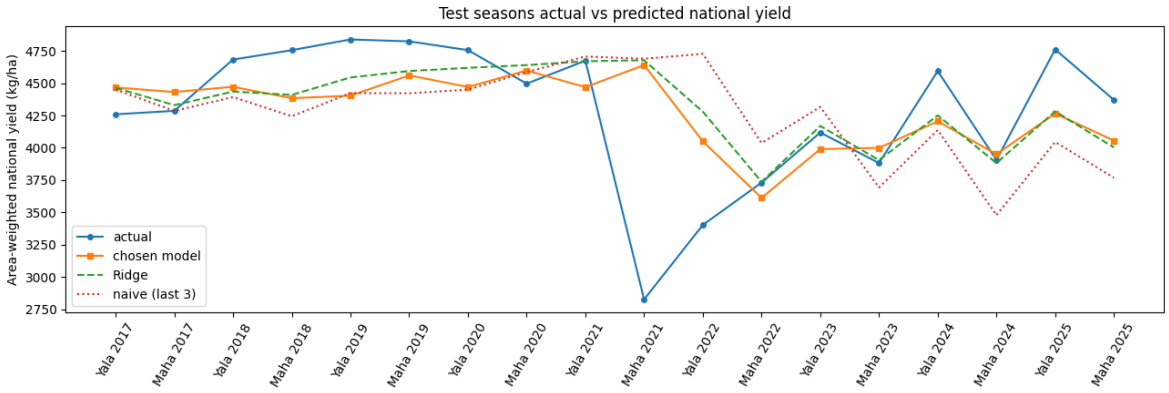



*Figure 3. Area-weighted national yield in the test seasons, actual and estimated. No model anticipated the 2021/22 Maha collapse.*

### 5.2 The 2021/22 fertilizer shock

For Maha 2021/22 the model estimated a normal season (approximately 4,640 kg/ha nationally) against an actual yield of 2,828 kg/ha, an average district error of 1,480 kg/ha. Excluding the two affected seasons, the test $R^2$ rises to 0.72. Errors were lowest in the large irrigated areas and in districts with stable yields (4-8%) and highest in the northern districts (14-24%), where most cultivation is rainfed and the historical record contains gaps from the conflict period.

### 5.3 Accuracy in planning terms

Table 3 expresses the test results in the terms a planner uses. Outside the fertilizer crisis, 93% of district estimates were within 20% of the official yield, and the implied national production was within 5.8% on average. The $R^2$ for production in tonnes is high mainly because larger districts produce more. It reflects the scale of the planning problem rather than a better model.

| Measure (test seasons 2017-2026) | All seasons | Excluding the two ban seasons |
|---|---|---|
| Accuracy (100% minus mean percentage error) | 87.6% | 90.6% |
| Median percentage error | 8.4% | 7.4% |
| District estimates within 20% of official yield | 86% | 93% |
| District production (tonnes), $R^2$ | 0.95 | 0.98 |
| National production per season, mean absolute error | 9.5% | 5.8% |

*Table 3. Accuracy of the selected model in planning terms. The naive three-season average placed 79% of district estimates within 20%.*

### 5.4 Incorporating government policy

The fertilizer policy cannot be learned from 1981-2016, because no shortage occurred in that period. It was therefore tested differently. Each model was trained on all seasons up to and including Maha 2021/22, and asked to estimate Yala 2022, the second shortage season. With a yes/no shortage input, the CatBoost error fell from 814 to 318 kg/ha and its national estimate (3,474 kg/ha) came within 71 kg/ha of the actual figure. The EBM overstated the effect, as it applied the full Maha 2021/22 collapse to a milder shortage. The deployed model includes this input as a scenario switch, which a planner can use when a shortage is expected. It was not used in model selection or in any reported test score.

### 5.5 Uncertainty of the estimates

The intervals produced by NGBoost were found to be overconfident. Its nominal 80% interval contained the actual yield in only 46% of test seasons. A conformal range [11] was therefore adopted, derived from the 80th percentile of validation errors (±545 kg/ha). It contained the actual yield in 72% of test seasons, and in 77% when the two fertilizer-ban seasons are excluded.

### 5.6 Flagging a poor harvest

Planners often need a simpler signal first, so each district-season was also classified as poor (more than 10% below its three-season average), normal or good. Each classifier was tuned on the validation years, and XGBoost was selected by macro F1 (0.465), narrowly ahead of logistic regression (0.460). On test, however, XGBoost reached a macro F1 of only 0.34 against 0.22 for a baseline that always predicts normal and identified 13% of poor harvests, whereas logistic regression reached 0.47 and identified 39%. Converting the regression estimate into a class identified 34% of poor harvests but was correct 71% of the time when it did, which makes it the most reliable warning. Tree-based classifiers performed poorly even after tuning, because poor seasons are rare in the training years.

Measures based on the predicted probabilities gave the same picture. The converted regression estimate ranked poor seasons best (ROC AUC 0.77, Matthews correlation coefficient [22] 0.26), followed by logistic regression (AUC 0.72, MCC 0.21), whereas the tree-based classifiers were close to chance (AUC 0.60-0.64, MCC 0.04-0.06). Because missing a poor harvest costs a planner more than a false alarm, the probability required to flag a poor harvest was chosen on the validation years by the F2 score, which weights recall twice as heavily as precision. For the selected XGBoost the best threshold was the lowest tried (0.20). It raised the share of poor harvests identified in the test seasons from 13% to 88%, including 43 of the 47 poor district-seasons in the fertilizer-ban period, but only by flagging three quarters of all seasons, with about a quarter of flags correct. A calibration curve showed that its probabilities are compressed into a narrow range and run high above about 0.25. A threshold on the regression estimate, which ranks poor seasons best, is therefore the more useful early warning.

## 6. Comparison with Existing Approaches

The stacked XGBoost approach of Senevirathne and Premasiri [5] and the support vector regression approach of Ekanayake et al. [6] were reproduced on the same data and time-based split. The stacked XGBoost achieved a strong validation result (345 kg/ha) but a test MAE of 451 kg/ha, comparable to the other tree ensembles. Support vector regression was the weakest model on validation (388 kg/ha) but competitive on test (443 kg/ha), consistent with the advantage of smooth models in the test period. As the original studies used different regions, periods and inputs, their published figures are not directly comparable, so reproducing the methods under identical conditions provides the appropriate comparison. The published support vector regression used the default kernel width. With the width tuned as well, its validation error fell to 345 kg/ha and its test error to 432 kg/ha, placing it within 1% of the best validation result. Its error varied more between years than the EBM's (standard deviation 83 against 79 kg/ha), so under the selection rule the EBM remained the chosen model. The choice rests on the validation years only, since selecting by test score would let the test seasons influence the choice, which is also why ridge regression (438 kg/ha on test) was not chosen.

## 7. Interpretation of the Model

The EBM is additive by construction. Each estimate is the sum of a baseline and one learned contribution per feature, together with a small number of pairwise terms. These contributions are its exact explanation and correspond to SHAP values [9]. SHAP, LIME [10] and the EBM's own feature functions present a consistent picture. Recent yield history is the dominant driver, followed by the irrigation mix and national conditions in the previous season. Among weather variables, the number of heavy-rain days has the clearest effect, with little impact up to about four days of 50 mm or more, followed by a progressively larger reduction. Very wet seasons, windy seasons and seasons much drier than normal also reduce the estimate. These attributions describe the model's behaviour and should not be read as causal effects.

Figure 4 shows this ranking across all 446 test seasons. The three-season average has the largest mean absolute SHAP value (248 kg/ha), more than twice that of last year's same season (116 kg/ha), followed by the previous season (106), the ARIMA forecast (100), the rainfed share (88) and the national previous season (83). Wind is the first weather variable, in ninth place (46 kg/ha), and the ten smallest features together add about as much as the three-season average alone. The colours show the direction. High recent yields raise the estimate by up to 700 kg/ha and low ones lower it by up to 600 kg/ha, while a high rainfed share, windy seasons and very wet seasons lower it.

This behaviour agrees with domain knowledge of paddy cultivation. Irrigated fields receive reliable water and therefore yield more than rainfed fields, and heavy rain near harvest causes flooding, lodging and disease, while strong monsoon winds flatten the crop. The learned curves in Figure 5 serve the same purpose as partial dependence plots, but are exact, because they are the model itself.

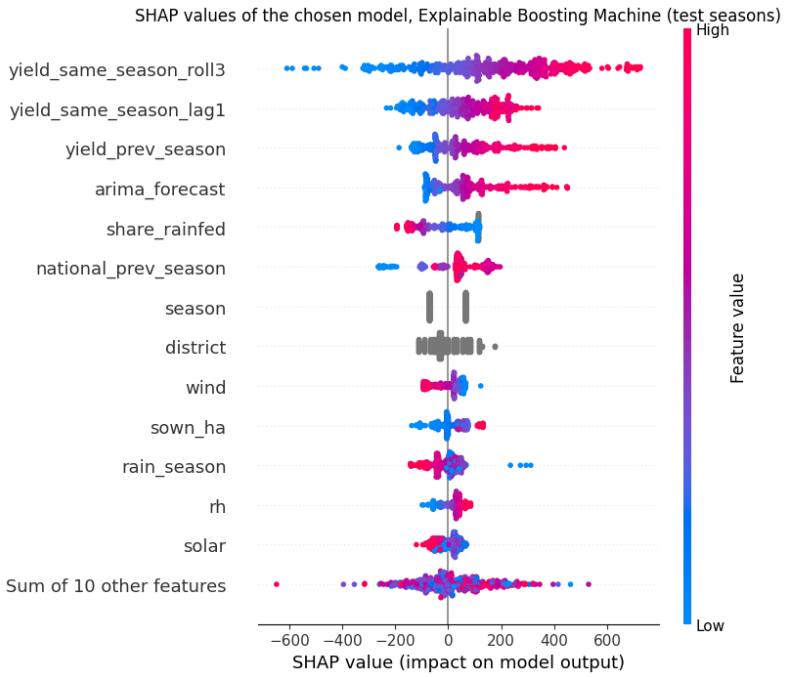



*Figure 4. SHAP values of the EBM for every test season, ordered by mean absolute contribution. Each point is one district-season, coloured by the value of the feature.*

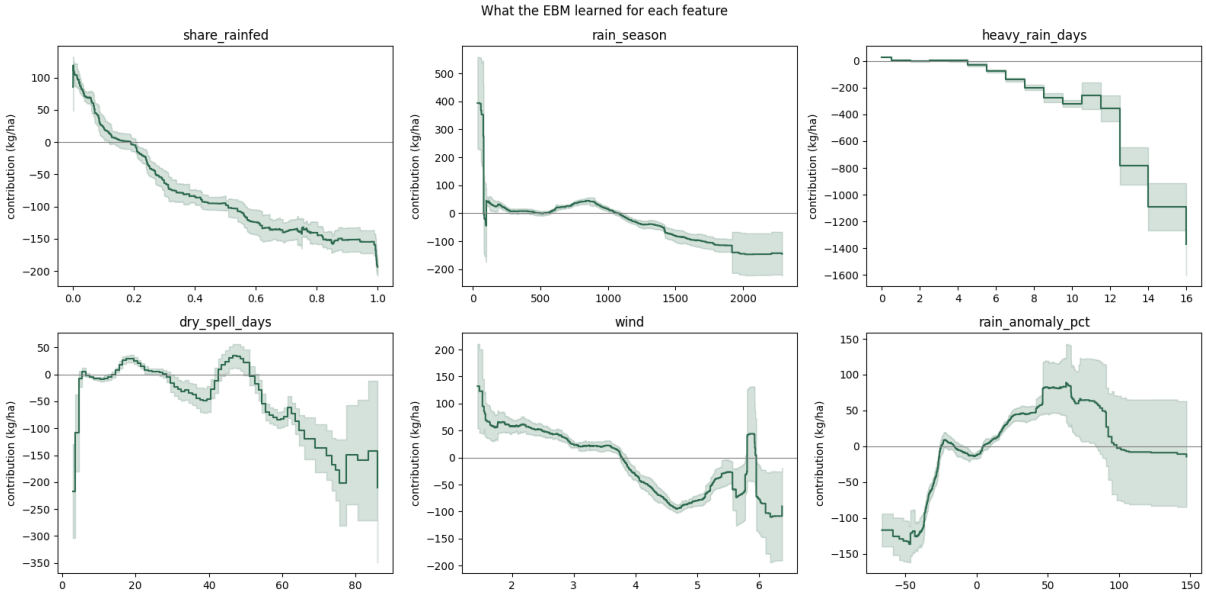



*Figure 5. Contributions learned by the EBM for the rainfed share and the weather features (kg/ha), with variation across bagged fits.*

SHAP dependence plots (Figure 6) quantify these effects. Each additional 100 kg/ha in a district's three-season average raises the estimate by approximately 31 kg/ha. Each additional ten percentage points of rainfed land lowers it by approximately 29 kg/ha, and seasons with eight or more heavy-rain days lower it by approximately 350 kg/ha. The explanation is also stable. When the model was retrained on random samples of 20% to 80% of the training seasons, the three-season average remained the leading feature in every case, and at least five of the six leading features were unchanged. The EBM's contributions add up exactly to each estimate (largest gap below 0.000001 kg/ha). LIME, rerun with ten random seeds and three kernel widths, returned the same five leading features with the same signs every time, although its local linear model explained only half of the variation around the season ($R^2$ 0.50). Permutation importance on the test seasons agreed moderately with the SHAP ranking (Spearman 0.57), with both placing the three-season average first. The differences arose where features overlap, as shuffling last year's yield costs little while the three-season average remains.

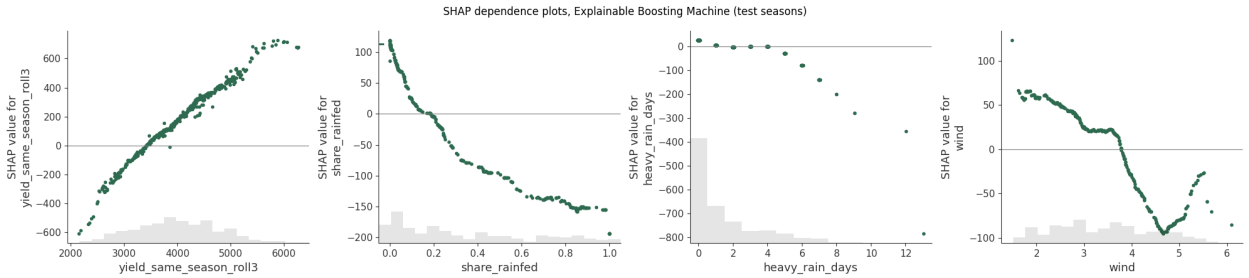



*Figure 6. SHAP dependence plots for the test seasons, showing the contribution of each feature against its value.*

For Polonnaruwa in Maha 2021/22 (actual 3,103 kg/ha, estimate 5,505 kg/ha), every contribution was positive, driven by the district's strong recent history. The explanation makes clear that the estimate rested on conditions that had held in every previous season and ceased to hold in 2021. For a decision maker, a large gap between the estimate and early field reports is therefore itself a warning sign.

Counterfactual explanations generated with DiCE [23] complement these attributions by asking what would have changed the estimate. Only the weather was allowed to vary, and only within the range each district had recorded in the same season type. For Galle in Yala 2023, a season correctly flagged as poor after 13 days of heavy rain, every counterfactual reduced the heavy-rain days to six or fewer, which raised the estimate to a normal harvest. For Polonnaruwa in Maha 2021/22 the reverse search found no recorded weather that could bring the estimate below 3,500 kg/ha, and the lowest estimate across 5,000 random weather combinations was 3,840 kg/ha against the actual 3,103 kg/ha. This confirms that the collapse lay outside the inputs available to the model.

## 8. Deployment

The selected model is provided through two components. An interactive Streamlit application allows a user to select a district and season, pre-fills each district's typical values, and returns the estimated yield, its likely range, a comparison with the district's history, the contribution of each input, a harvest outlook, and a fertilizer-policy scenario. Further pages vary one input across its range, map and rank district yields for any season, compare the models in three views (validation, test and the published benchmark) above the SHAP values of the test seasons, and estimate a whole CSV file of district-seasons. The application does not run the model itself. It sends each request to a FastAPI prediction service, which validates the input against a data contract, builds the features from the stored history, returns the estimate with its explanation through the */predict* endpoint (or many estimates through */predict/batch*), and records every prediction in a structured log. Both components are packaged as Docker containers and deployed to Azure Container Apps, and the application falls back to a local copy of the model if the service cannot be reached. On the held-out Polonnaruwa Maha 2025/26 season, the model estimated 5,026 kg/ha against an actual record yield of 6,095 kg/ha, again underestimating a sharp departure from recent history. Under the shortage scenario it would estimate 3,068 kg/ha.

The deployed web application is available at paddy-yield-app.grayflower-d27356cf.malaysiawest.azurecontainerapps.io, the prediction API at paddy-yield-api.grayflower-d27356cf.malaysiawest.azurecontainerapps.io/docs, and the complete source code, data and deployment workflows in the GitHub repository at github.com/Uthpalagit/paddy-yield-prediction-269219J. A backup copy of the application, which runs the same model file without the API, is hosted on Streamlit Community Cloud at paddy-yield-prediction-269219j-2wfd9iajbb7ebf2jpuzzgb.streamlit.app, so the application remains available if the Azure deployment is stopped.

Following standard MLOps practice, a continuous-integration workflow runs automated tests of the service and builds both containers on every change, and a scheduled monitoring job compares new inputs with the training data and, once official yields are released, the error with the validation level. Applied retrospectively to 2017-2026, the standard drift score (PSI) proved unsuitable season by season, because all districts share the

same weather. The informative signal was the share of districts whose recent yields lay outside their training range, which exceeded half from Maha 2019/20 to Yala 2022, the period in which the model lagged. Error alerts fired in four seasons, and a champion-challenger rule prevented one inferior retrained model from being deployed, although retraining alone did not improve on the original model.

Experiments and models are tracked with MLflow [24]. Every model in the comparison is logged as a run with its settings and scores, and each trained model is stored as a version in the model registry. When the champion-challenger rule was replayed through the registry, the quality gate rejected the challengers before Maha 2018/19 and Yala 2022 and promoted two others, and the deployed model carries the production alias together with the fingerprints of the data and model file it was built from. The same fingerprints are recorded by DVC [25], whose pipeline reruns the monitoring check whenever the raw data, the model or the monitoring code changes, so every estimate can be traced to the exact data and model that produced it.

Some practices from the MLOps workshop were applied only in part. The DVC pipeline reruns the monitoring check but does not retrain the model, and retraining remains a manual step through the notebook. The CI workflow runs the tests and builds both containers but has no lint step or automated quality gate, and deployment replaces the running version directly, using stored credentials rather than keyless sign-in, without an approval step, canary release or scripted rollback. These would be needed before use by an agency.

## 9. Critical Discussion

The main limitation is that the model learns only from conditions that occurred in the past. It did not anticipate the 2021/22 fertilizer collapse, and its tree-based terms cannot predict beyond the yields seen in training, so record seasons are underestimated. The shortage effect rests on only two seasons, and the conformal range contained the actual yield in 72% of test seasons rather than the intended 80%. The past is also a moving reference. From Maha 2019/20 to Yala 2022 more than half of the districts had recent yields outside their training range, and a changing climate may make such periods more frequent.

The evaluation has limits of its own. Every model was tested on one period, the 446 district-seasons of 2017-2026, which contains both record yields and the fertilizer ban, so the order of models that are close in accuracy could change in another decade. The validation years also informed many design decisions, and this reuse had a measurable cost. Growth-stage weather and the consensus of three feature selection methods looked better on validation but not on test (447 and 464 kg/ha against 446), which is why they were reported but not adopted. Finally, SHAP and LIME describe how the model uses its inputs, not how weather or irrigation cause yield.

The data has quality issues as well. Survey records are missing for parts of the conflict period, the Mahaweli System H and Udawalawe series end in 2018, and temperature, humidity and wind come from a coarse grid that gives neighbouring districts the same values. Twenty-one war-year yields were recorded as zero and had to be treated as missing. There is also no district-level record of fertilizer applied, seed varieties, pests or reservoir releases, so part of the variation in yield cannot be explained by any available input.

These gaps create a risk of unequal accuracy. The average error was 4-8% in the large irrigated districts but 14-24% in Kilinochchi, Mullaitivu, Mannar and Jaffna, which are mostly rainfed and have the most gaps. The gaps are not random, because they follow the years of fighting, so the districts most affected by the war are also the ones the model serves least well. If decisions on support or procurement relied on the model alone, farmers who already have the least irrigation would receive the least reliable estimates. Estimates for these districts should therefore carry a wider margin and be checked against field reports, and accuracy should be reported district by district rather than only as a national average.

In practice, an early estimate could help the Department of Agriculture and the Paddy Marketing Board act several months sooner. The model should support the official survey rather than replace it. An underestimate could lead to unnecessary imports that lower farm-gate prices, and an overestimate could delay support after a poor harvest. For this reason each estimate is shown with its range and its explanation, and the application states

its limitations. The fertilizer-shortage switch produces a scenario learned from two seasons and should not be read as a measured effect of any future policy.

The data itself raises few ethical concerns. It is public and aggregated by district, with no personal or household information, so no consent was required (Section 2). The greater risk lies in how the estimates are used. A single precise figure can be trusted more than it deserves, so the final decision should remain with officials, who can see the range and the contribution of each input because the EBM is a glass-box model. Every estimate from the deployed service is logged together with the data and model that produced it, so any decision based on it can be traced and reviewed.

## 10. Conclusions and Recommendations

The project demonstrates that district paddy yields can be estimated at the end of each season with an accuracy of approximately 88%, several months before official figures are available, and that each estimate can be explained in terms a planner can verify. The analysis also shows that most of the predictive signal lies in yield history and irrigation, that improvements in data quality matter more than the choice among strong algorithms, and that a policy shock such as the 2021 fertilizer ban cannot be learned from historical data alone, although once observed it can be represented as a policy input.

The limitations discussed in Section 9 point to the following recommended steps.

- Incorporate district-level fertilizer distribution data to capture input shocks and to measure their effect more precisely.
- Add reservoir storage at the start of each Yala season, if the irrigation authorities release it.
- Evaluate finer satellite imagery (such as Sentinel-2 at 10 m) over mapped paddy fields, once enough seasons exist for training.
- Evaluate a combination of the EBM with a smooth model to improve extrapolation in exceptional seasons.
- Confirm the benefit of the qualitative features, especially the El Niño phase, on the 2026/27 seasons once they are published.
- Evaluate a smaller learning rate for the EBM when it is retrained on the 2026/27 seasons.
- Operate the monitoring each season against the official DCS figures, treating a high share of districts beyond their training range as a signal to verify the estimates against field reports.

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